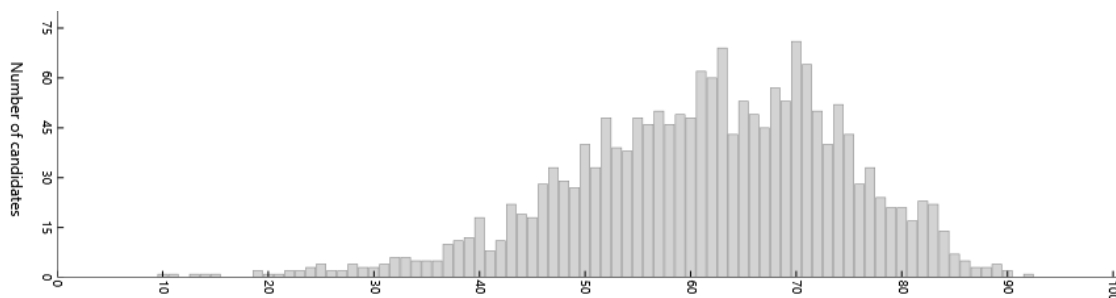




2018 ATAR course examination report: Biology

Year	Number who sat	Number of absentees
2018	1835	29
2017	1810	20
2016	1822	30

Examination score distribution–Written



Summary

The examination paper had three sections. Section One comprised 30 multiple-choice questions, Section Two comprised five short answer questions and Section Three comprised two extended answer questions from Unit 3 and two from Unit 4. Candidates were required to answer all questions in Sections One and Two, and one question from each of Units 3 and 4 in Section Three. The examination was attempted by 1835 candidates and had a mean score of 61.37%. Scores ranged from a minimum of 10% to a maximum of 91.75%.

Attempted by 1835 candidates Mean 61.37% Max 91.75% Min 10.00%

Section means were:

Section One: Multiple-choice	Mean 76.63%		
Attempted by 1835 candidates	Mean 22.99(/30)	Max 30.00	Min 10.00
Section Two: Short answer	Mean 57.82%		
Attempted by 1833 candidates	Mean 28.91(/50)	Max 45.25	Min 0.25
Section Three: Extended answer Unit 3	Mean 42.39%		
Attempted by 1796 candidates	Mean 4.24(/10)	Max 10.00	Min 0.00
Section Three: Extended answer Unit 4	Mean 54.14%		
Attempted by 1814 candidates	Mean 5.41(/10)	Max 10.00	Min 0.00

General comments

The majority of candidates were well-prepared and attempted all questions.

The overall mean scores for the 2018 and 2017 examinations were very similar (61.37% versus 61.42% respectively). In 2018 compared to 2017, candidates did slightly better in the multiple-choice section (76.63% versus 71.25%) and the extended answer in Unit 4 (54.14% versus 47.02%) and slightly worse in the short answer section (57.82% versus 60.16%) and the extended answer in Unit 3 (42.39% versus 53.93%). The paper discriminated between candidates and the standard deviations ranged from 11.69% for the multiple-choice questions to 18.20% and 18.28% for the extended answer questions in Units 3 and 4, respectively. It is noted that statistics show that there is difficulty in discriminating between the more able candidates. Perhaps a different way of presenting questions and the related marking key should be investigated in the extended answer section.

Advice for candidates

- Include units with quantitative data (e.g. when reporting data from a graph or table).
- Read the question carefully and make sure that you answer the question asked. Take particular care with the extended answer questions in Section Three which often have subtle components that require attention.
- Use formal and precise language and scientific terminology in answering questions.
- Communicate clearly in your written answers.

Advice for teachers

- Instruct students how to answer questions according to the 'instruction' verb in the question (e.g. state, define, discuss, and explain).
- Give students practice at decoding questions.
- Prepare students to provide coherent answers to extended answer questions in Section Three (e.g. where information is logically organised, connects different points and demonstrates an understanding the key concepts).

Comments on specific sections and questions

Candidates answered the short answer and especially the multiple-choice questions well but had more difficulty with the extended answer questions in Section Three.

Section One: Multiple-choice (30 Marks)

Most candidates performed well in this section (mean score 76.63%). All candidates attempted every question. Although most candidates got most questions correct, no single question was answered correctly by all candidates.

Eight questions (1, 7, 10, 14, 16, 20, 21 and 23) had mean scores of 90% or above. Most of these questions tested factual recall or required candidates to obtain data from a figure but two (16 and 20) required an application of knowledge.

Three questions (3, 6 and 8) had mean scores below 50%. Question three required candidates to distinguish between a scientific observation versus predictions and inferences. Question six required candidates to realise that both fungal and bacterial cells have cell walls. Question eight required candidates to understand that a terrestrial bird will lose most water by breathing.

Questions 14 and 15 required candidates to obtain data from a phylogenetic tree. Candidates performed well in question 14 (mean score of 94%), suggesting that they understood how to interpret such trees. The mean score for question 15 was, however, only 54%. It appears that many candidates did not inspect the tree carefully enough to realise that, while cycads, ginkgoes and conifers formed a single related group of plants, cycads, ginkgoes and mosses did not.

Question 1 attempted by 1835 candidates	Mean 0.98(/1)	Max 1.00	Min 0.00
Question 2 attempted by 1835 candidates	Mean 0.87(/1)	Max 1.00	Min 0.00
Question 3 attempted by 1835 candidates	Mean 0.39(/1)	Max 1.00	Min 0.00
Question 4 attempted by 1835 candidates	Mean 0.73(/1)	Max 1.00	Min 0.00
Question 5 attempted by 1835 candidates	Mean 0.89(/1)	Max 1.00	Min 0.00
Question 6 attempted by 1835 candidates	Mean 0.44(/1)	Max 1.00	Min 0.00

Question 7 attempted by 1835 candidates	Mean 0.90(/1)	Max 1.00	Min 0.00
Question 8 attempted by 1835 candidates	Mean 0.43(/1)	Max 1.00	Min 0.00
Question 9 attempted by 1835 candidates	Mean 0.80(/1)	Max 1.00	Min 0.00
Question 10 attempted by 1835 candidates	Mean 0.90(/1)	Max 1.00	Min 0.00
Question 11 attempted by 1835 candidates	Mean 0.53(/1)	Max 1.00	Min 0.00
Question 12 attempted by 1835 candidates	Mean 0.79(/1)	Max 1.00	Min 0.00
Question 13 attempted by 1835 candidates	Mean 0.85(/1)	Max 1.00	Min 0.00
Question 14 attempted by 1835 candidates	Mean 0.94(/1)	Max 1.00	Min 0.00
Question 15 attempted by 1835 candidates	Mean 0.54(/1)	Max 1.00	Min 0.00
Question 16 attempted by 1835 candidates	Mean 0.91(/1)	Max 1.00	Min 0.00
Question 17 attempted by 1835 candidates	Mean 0.79(/1)	Max 1.00	Min 0.00
Question 18 attempted by 1835 candidates	Mean 0.73(/1)	Max 1.00	Min 0.00
Question 19 attempted by 1835 candidates	Mean 0.82(/1)	Max 1.00	Min 0.00
Question 20 attempted by 1835 candidates	Mean 0.91(/1)	Max 1.00	Min 0.00
Question 21 attempted by 1835 candidates	Mean 0.98(/1)	Max 1.00	Min 0.00
Question 22 attempted by 1835 candidates	Mean 0.61(/1)	Max 1.00	Min 0.00
Question 23 attempted by 1835 candidates	Mean 0.91(/1)	Max 1.00	Min 0.00
Question 24 attempted by 1835 candidates	Mean 0.87(/1)	Max 1.00	Min 0.00
Question 25 attempted by 1835 candidates	Mean 0.87(/1)	Max 1.00	Min 0.00
Question 26 attempted by 1835 candidates	Mean 0.71(/1)	Max 1.00	Min 0.00
Question 27 attempted by 1835 candidates	Mean 0.88(/1)	Max 1.00	Min 0.00
Question 28 attempted by 1835 candidates	Mean 0.80(/1)	Max 1.00	Min 0.00
Question 29 attempted by 1835 candidates	Mean 0.69(/1)	Max 1.00	Min 0.00
Question 30 attempted by 1835 candidates	Mean 0.54(/1)	Max 1.00	Min 0.00

Section Two: Short answer (100 Marks)

This section was attempted by almost all candidates. Although no one candidate got full marks for the entire section (maximum score was 90.5%), at least one candidate achieved full marks for every part of every question. There was also at least one candidate achieving full marks for each of Questions 31, 33 and 35; the maximum scores for Questions 32 and 34 were 95% and 97.5% respectively. The mean scores per question ranged from 66.65% (question 31) to 45.35% (question 32). The mean scores for different parts of the same question tended to vary.

Question 31 attempted by 1832 candidates Mean 13.33(/20) Max 20.00 Min 0.00

This question covered topics in the 'homeostasis' area of the syllabus. Part (a), which asked candidates to add labels to a diagram of stimulus-response model, and part (c), which asked candidates to identify the most toxic nitrogenous waste, were well completed generally and had the highest mean scores (86.25 and 91% respectively). Candidates struggled most with part (e) with a mean score of 49.5%, which required candidates to explain why marine bony fish excrete only a small volume of urine. Candidates often focused on the excretion of ammonia or produced an indiscriminate list of points relating to osmoregulation in a marine bony fish rather than explaining why little urine is produced. Part (b) asked candidates to identify the type of nitrogenous waste produced by different organisms. A surprising number of candidates indicated that a desert rat produces uric acid and/or that a river dolphin produces ammonia, i.e. failed to understand the importance of taxonomic position in regard to the type of waste an animal produces.

Question 32 attempted by 1828 candidates Mean 9.07(/20) Max 19.00 Min 0.00

This question covered topics in the 'continuity of life on earth' area of the syllabus. Candidates performed best in part (a), which asked about the diploid number of chromosomes in a mule, although the mean score was only 64%. While candidates typically got the number of chromosomes correct, many failed to include a sufficient explanation of their answer and/or to use specific terms such as 'haploid', 'sperm', 'egg' or 'gamete'. On average, candidates performed poorly in part (b) (mean score 37.5%) and part (e) (mean score 25.75%). Part (b) asked candidates to explain why mules could not produce functional gametes. While many candidates mentioned this related to the odd number of chromosomes present in the mule, most struggled to articulate the consequences of this for gamete production and few even mentioned meiosis. In part (e), many candidates failed to recognise that the evolution of forefeet in horses is an example macroevolution (major or large-scale evolutionary scale) and not microevolution. Part (d) asked candidates to explain how biologists know about the evolution of forefeet in horses. Most candidates recognised that this was via fossils but many went on to provide a rehearsed answer about fossilisation in general rather than applying their knowledge to answer the specific question about horse forefeet (mean score was only 48.5%).

Question 33 attempted by 1828 candidates Mean 11.93(/20) Max 20.00 Min 0.00

This question covered topics in the 'heredity' area of the syllabus. Candidates generally did well with parts (c)(i) (mean score 92%) and (c)(ii) (mean score 73.75%), which asked candidates to determine the probability of obtaining an albino offspring from a particular cross and to draw a Punnett square to show how they obtained this probability. The mean scores for part (e) (which asked about how biotechnology could be used to determine the father of a litter of guinea pigs) and part (a) (which asked candidates to explain in words how non-albino guinea pigs could produce albino offspring) were also generally done well (mean scores of 72.75% and 71.67%, respectively). On the other hand, candidates struggled to articulate why albinism cannot be a sex-linked trait in guinea pigs in part (b) (mean score of 36.75%) and why albino guinea pigs are common in captive populations but rare in the wild (part (d), 41.5%). In relation to part (d), candidates often talked about inbreeding and genetic drift rather than focusing on artificial and natural selection.

Question 34 attempted by 1825 candidates Mean 11.92(/20) Max 19.5 Min 0.00
This question covered topics in the 'science inquiry skills' area of the syllabus. Part (a), preparation of a graph, was generally very well completed and had a high mean score (81.5%). Nevertheless, a number of candidates made fundamental errors such as presenting poor titles, a bar graph, reversed axes and/or inaccurate plots. The mean score for part (b)(i), which required candidates to estimate data for barley from the graph was also high (82%) and higher than part (b)(ii) which required candidates to estimate data for lupins in a part of the graph where there was a marked change between the two surrounding data points.

A number of candidates forgot to include the units for these questions. Part (e) asked candidates to deduce why barley plants are more tolerant of salt than lupins and had the lowest mean score at only 36.25%. Many candidates noted the differences in xylem salt concentration between the two types of plant but did not go on to deduce how these differences might be achieved. Candidates often confused the concepts of reliable and valid data in part (c) (mean score 46.33%) and part (d) (mean score 50%) and some candidates talked about growing plants under identical conditions to make a 'fair' experiment rather than using meaningful terminology.

Question 35 attempted by 1827 candidates Mean 11.72(/20) Max 20.00 Min 0.00
This question covered topics in the 'infectious diseases' area of the syllabus. Part (b), which asked candidates to describe how malaria is transmitted from an infected person to an uninfected person, was well done and had the highest mean score (76.75%). Part (a) asked about the structural features of protists and was seemingly a straightforward question requiring only factual recall to obtain full marks. However, this part had the lowest mean score in this section at only 46.75%. Many candidates mistakenly identified protists as prokaryotes.

Section Three: Extended answer Unit 3 (20 Marks)

The mean score for the questions in Unit 3 in Section Three of the paper (42.4%) was the lowest for any part of the paper. The number of candidates who attempted this part of the examination was also the lowest, although most (98%) candidates nevertheless made an attempt. Question 37 was less popular than Question 36 and had a lower mean score (38.15% compared to 44.9%).

Question 36 attempted by 1132 candidates Mean 8.98(/20) Max 20.00 Min 0.00
The mean scores for parts (a) and (b) of question 36 were both below 50% (48.1 and 42.3% respectively). Nevertheless, the maximum possible score (10/10) was achieved in both parts. Part (a) asked candidates to discuss the likely consequences of the deletion of one nucleotide from the DNA in a cell on protein production and on the structure and function of the cell. Many candidates discussed the consequences for protein production but did not address how the mutation would affect cell structure and function and often merely restated that the structure and function would change. It should be emphasised that restating the question does not gain marks and that all parts of a question need to be answered to obtain a good score. Some candidates repeated rehearsed answers about types of mutation or provided an indiscriminate list of details about transcription and translation per se rather than focussing their answer on the question asked. Part (b) asked candidates to discuss how 'genetic drift' and 'gene flow' change allele frequencies in the gene pool of a population. Many answers referred to genetic drift being the same as gene flow, i.e. failed to recognise that genetic drift is about random changes in allele frequencies within a population and not about alleles being exchanged between different populations. Other answers described bottleneck and founder effects in detail but did not display an understanding of which type of the two changes these effects represent. Candidates frequently stated that genetic drift is random, but then displayed misunderstanding by going on to discuss how it selects characteristics favourable to survival.

Question 37 attempted by 664 candidates Mean 7.63(/20) Max 19.50 Min 0.00
 The mean scores for parts (a) and (b) (36.1% and 41.7% respectively) were lower than those for Question 36. The maximum score was 10/10 in part (a) and 9.5/10 in part (b), thus high scores were possible despite the low means. Part (a) asked candidates to discuss how environmental factors can cause mutations in genes and how errors in meiosis can cause mutations in chromosome numbers. Few candidates were able to provide an in-depth answer to this seemingly straightforward question. Few explained how environmental factors actually change DNA structure. Simple repetition of the information in the question, e.g. 'environmental factors cause mutations in genes' did not score marks. Many candidates confused epigenetic change (heritable but reversible changes in DNA packaging/gene expression) with mutation (heritable and permanent changes to DNA sequence). Many candidates stated that natural selection or environmental change were causes of mutation 'because species need to survive', thus demonstrating a fundamental misunderstanding of the process of microevolution. Part (b) asked candidates to discuss why populations with reduced genetic diversity face an increased risk of extinction and how biotechnology can be used to reduce this risk. Most candidates showed a good understanding of the risks of decreased diversity but struggled with the biotechnology aspect of the question. Many wrote detailed descriptions of how a transgenic organism is made, presumably from study notes or pre-prepared answers, but failed to address the question of how biotechnology could be used to increase diversity.

Section Three: Extended answer Unit 4 (20 Marks)

Overall candidates performed better in Unit 4 (mean score 54.1%) than Unit 3 (mean score 42.3%). Almost all (99%) candidates attempted this part of the examination. Question 38 was more popular than question 39 but had a lower mean score (50.15% compared to 62.2%).

Question 38 attempted by 1212 candidates Mean 10.03(/20) Max 19.50 Min 0.00
 The mean scores for parts (a) and (b) were both close to 50%. The maximum possible score (10/10) was achieved in both parts (a) and (b). Part (a) asked candidates to discuss one structural and one physiological process that enable mammals living in cold environments to maintain a constant core body temperature. Many candidates wrote non-specific answers about fur being 'like a blanket' and 'trapping heat' but did not discuss the role of trapped air as an insulator. Hibernation was mentioned in some answers but not awarded marks as, although a physiological process, it does not maintain a constant core body temperature. It is emphasised that candidates should read carefully and answer the question asked. Part (b) asked candidates to discuss how population density and the provision of healthcare can influence the susceptibility of an urban area to an influenza epidemic. Strong answers paid attention to both parts of the question. A population's 'susceptibility to infection' and the 'rate of infection' are different and are relevant aspects of a sound answer. Many candidates struggled to envisage what constitutes an 'urban' area.

Question 39 attempted by 602 candidates Mean 12.44(/20) Max 20.00 Min 0.00
 Overall candidates performed well in both part (a) (mean score 65.6%) and part (b) (mean score 59.7%) of this question relative to the other questions in Section Three. The maximum possible score (10/10) was achieved in both parts (a) and (b). Part (a) asked candidates about how phytophthora dieback disease spreads and the management strategies that can be used to control this spread. Candidates' knowledge of phytophthora dieback disease was good generally and most were easily able to discuss management strategies. A significant number of candidates wrote that dieback disease is due to a fungal infection. Candidates are encouraged to make sure that their knowledge is current. Part (b) asked candidates about how a xerophyte minimises water loss while still allowing for gas exchange. Many candidates wrote a pre-prepared list of all xerophyte adaptations with little regard to the question posed. Answers that paid attention to how gas exchange is enabled while water is

conserved scored more highly. This is a classic example of where candidates are encouraged to explore the difference between the two parts of the extended answer questions (i.e. part (a) and part (b) of each question). Part (b) questions often have subtle aspects that require attention, or candidates are asked to apply their understanding in more advanced, complex, and unfamiliar contexts.